

Structural Lottery Tickets: Node Pruning for Efficient Neural Networks

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I. INTRODUCTION

The lottery ticket hypothesis shows that sparse subnetworks exist within dense models that can match full network performance from their original initializations. The standard approach currently uses weight masking, which sets pruned weights to zero. This leaves matrix dimensions unchanged, and provides no practical speedup on commodity hardware, since sparse tensor operations lack native acceleration. Structural pruning addresses this by removing entire neurons. Identifying a structural lottery ticket will speed up training, latency, and use up less memory.

II. RESEARCH METHODS

A. *LENET on MNIST*

For the MNIST experiments we use a fully connected network with architecture: 784 - 300 - 100 - 10, following a Lenet structure. The network uses ReLU activations, a cross-entropy loss optimized with Adam at a learning rate of $1.2e-3$, and a batch size of 60. MNIST consists of 60,000 training and 10,000 test grayscale images of handwritten digits 0-9. We hold out 5,000 samples from the training set for validation to use for early stopping. The test set is not touched until final accuracy is calculated.

We apply iterative structural pruning over 15 rounds with a pruning rate of 20 percent per round. At each round the model is trained with early stopping monitored on the validation set, with a patience of 5 checks evaluated every 100 gradient steps and a maximum of 50,000 iterations. After training, all nodes are ranked by their importance score, which we define as the mean absolute value of their outgoing weights. The 20 percent least important nodes are removed structurally, deleting the row from the prior weight matrix and the column from the following weight matrix, with the constraint that no layer is reduced below one node. The surviving weights are then reset to their original random initialization values and the reduced network is retrained and the process is repeated.

We train a control model in parallel at each round with the same pruned architecture, but random weights. This serves as our random reinitialization baseline. All experiments are repeated across three independent runs and results are averaged.

B. *CONV-2 on CIFAR 10*

The iterative structured pruning procedure was expanded onto a two layer convolutional neural network referred to as the CONV-2 model. The architecture of this model consists of two convolutional layers, each with 64 filters, followed by pooling after the second layer. It then proceeds into fully connected layers with architecture of 256-256-10 for classification. In this procedure, we used the cross-entropy-loss

with Adam optimization with a learning of $2e-4$. Instead of using the MNIST dataset on this CONV-2 model, it was tested using the CIFAR-10 dataset. The CIFAR-10 data set is a more complicated data set as it consists of 60,000 images, of which 45,000 were used for training, 5,000 for the validation set, and 10,000 for the test set. These images consist of 32x32 color images that are set up to be classified as one of 10 potential different classes. This expansion into a more complicated dataset and neural network architecture would help analyze if existing structured lottery tickets exist under more difficult conditions and data.

In this model and data set, we applied a very similar procedure of iterative structural pruning from LENET to the CONV-2 model. The difference is that instead, at every round, each layer pruned a certain percentage of filters and neurons instead of being pruned globally.

The procedure of the experiment is as follows, first we construct the CONV-2 model and save the initializations of the weights as the original values. After saving these weights, we run the procedure. It takes in the current CONV-2 pruned model and restores the original initializations for the surviving filters and neurons. This model is referred to as the CONV-2 Winning-Ticket model. Then it constructs another CONV-2 model that is of the same size and architecture as the Winning-Ticket model, but instead it is initialized with random weights. This model is referred to as the CONV-2 Random-Reinitialization model. Both models are trained in parallel following the same early stopping method that was used in LENET, but with a maximum iteration of 10,000. After training, both models are evaluated on the testing data set and compared to each other. Finally, we take the Winning-Ticket model, with its new trained weights, and apply the layer-wise structural pruning algorithm on it. It was calculated using the same metric as in LENET, by calculating the mean absolute value of their outgoing weights. In this experiment, we pruned five percent of the filters at each convolutional layer, and ten percent of neurons at each fully connected layer except the output layer. Then we took this pruned model and repeated the procedure. As presented in the LENET procedure, this was also ran three times, and the results at each size of the pruned Conv-2 model were averaged.

C. *CONV-4 on CIFAR 10*

We extend the iterative structured pruning procedure to a four-layer convolutional model (CONV-4) on CIFAR-10. The CONV-4 architecture contains two convolutional blocks: 64, 64, pool, then 128, 128, pool, followed by fully connected layers 256-256-10. CIFAR-10 was split into 45,000 training images, 5,000 validation images, and 10,000 test images, with

validation used only for model selection and early-stopping iteration tracking.

The optimization setup followed Adam with learning rate 3×10^{-4} , cross-entropy loss, and iterative training with validation checks every 100 updates. At each pruning round, we compared two models of identical architecture: (1) a Winning Ticket (WT) model reset to its original initialization for surviving units, and (2) a Random Reinitialization (RR) baseline with newly sampled random weights.

For pruning, we intentionally used a hybrid strategy to compare prior team methods in a deeper network: convolutional layers were pruned using the CONV-2 layer-wise filter-magnitude method, while fully connected hidden layers used the LENET-style global ranking method across FC hidden neurons. Convolutional layers were pruned at 5 percent per layer each round, and fully connected hidden layers were pruned by removing the bottom 10 percent globally by outgoing-weight magnitude score. After each round, surviving architecture indices were tracked, the model was rebuilt at reduced width, and the iterative train-prune-reset cycle was repeated until minimum layer-size constraints were reached.

TABLE I
NETWORK ARCHITECTURES AND TRAINING PARAMETERS

Network	Lenet	Conv-2	Conv-4
Convolutions	—	64, 64, pool	64, 64, pool, 128, 128, pool
FC Layers	300, 100, 10	256, 256, 10	256, 256, 10
All/Conv Weights	266K	4.3M / 38K	2.4M / 260K
Iterations/Batch	50K / 60	10K / 128	5K / 60
Optimizer	Adam 1.2×10^{-3}	Adam 2×10^{-4}	Adam 3×10^{-4}
Pruning Rate	fc 20%	conv 5%, fc 10%	conv 5%, fc 10%

III. EVALUATION

A. LENET on MNIST

We recorded accuracy and the best early stopping iteration across 15 rounds of iterative structural pruning, reducing the network from 100% of nodes down to 4.4%. We averaged the results from 3 independent trials.

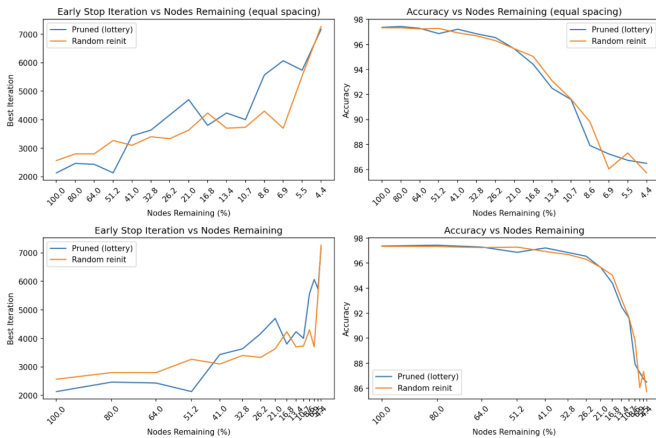


Fig. 1. LENET Results

The models start at 97.5% test accuracy and remain competitive in early pruning rounds. Accuracy holds for both models down to 26% of nodes remaining where. Below that, accuracy drops off fast for both. Where they both report accuracy of mid to high 80s in the final rounds. The lottery ticket model maintains a marginal accuracy advantage over the random baseline across most pruning levels, though the gap is narrow and the two lines largely track each other.

The early stopping iteration metric shows a clearer separation in the early rounds. In the early rounds, The lottery ticket model converges in 2,100 iterations at full capacity compared to the 2,600 for the random baseline. This suggests the rewind initialization provides a more favorable starting point. However, as pruning progresses both models require increasing iterations to converge, and the gap narrows. At extreme pruning levels both models spike sharply in iterations required. The random baseline occasionally converges faster than the lottery ticket model in the final rounds.

Overall the results are inconclusive. Even though there is a modest and consistent early advantage for the lottery ticket model in training efficiency, the separation is not large enough to draw a strong conclusion.

B. CONV-2 on CIFAR-10

For the CONV-2 procedure, the number of training iterations, training, and testing accuracy were recorded. In the following graphs, we compared the metrics of the Winning-Ticket and the Random-Reinitialized model at every stage of its pruned architecture. In addition, we present two graphs from the same results, but with different values on the x-axis to represent the size of the model. The x-axis is the number of filters and neurons in the current pruned model for each layer, or the percentage of weights the pruned model had in relation to the 4.3 million parameters of the base model.

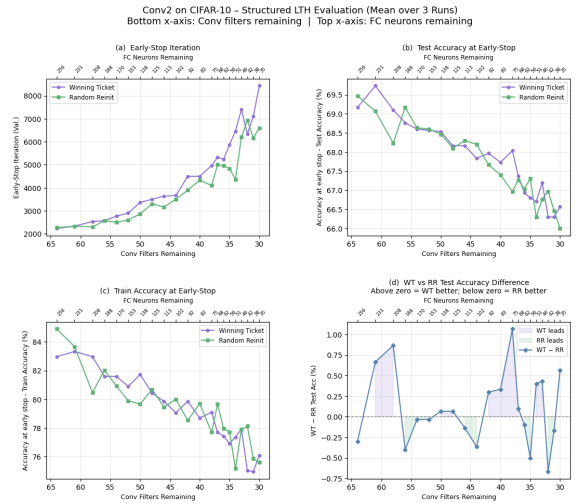


Fig. 2. CONV2 Results

The baseline test accuracy of the CONV-2 model was averaged to be around 69.2 percent. It is then shown that through many rounds of layer-wise iterative structured pruning, the

testing accuracy stayed in the range from 69 to 68 percent for both the Winning-Ticket and Random-Reinitialization models. This period occurred when the size of the model decreased from 64 filters to 38 filters and in relation to the number of parameters, it was from 100 through 20 percent of its original size. The precision of the CONV-2 structured winning

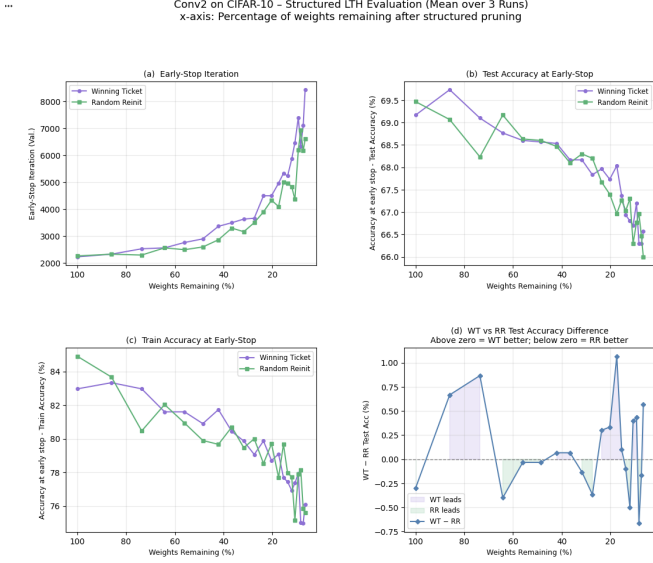


Fig. 3. CONV2 Results

ticket model, despite never reaching the same accuracy as the baseline model, was still close, even as the size decreased significantly. Yet, one important aspect from the results is that the testing accuracy between the Winning ticket and Random Reinitialized models oscillated. There was no clear consistent advantage of the Winning Ticket model performing better. At times, though briefly, as seen in the WT-RR graph, the Random Reinitialized model did perform as well or just a bit better than the WT model. This is different compared to the dominance of the WT model’s accuracy presented in unstructured pruning from the LTH paper.

In addition, another aspect in relation to the unstructured Winning Lottery Ticket findings, is that for this structured Winning Lottery Ticket it generally needed many more training iterations. As the size of the pruned CONV-2 model decreased, many more iterations were needed for the WT and RR model to train. Thus, it was not able to replicate the characteristics that such Winning Lottery Tickets could be trained in the same if not even fewer iterations than the baseline CONV-2 model.

C. CONV-4 on CIFAR-10

For the CONV-4 experiment we extended iterative structured pruning to a four convolutional layer network with two pooling stages and a 256-256-10 fully connected classifier. We tracked the same metrics as in the LENET and CONV-2 sections at each pruning round: (1) the early-stop iteration defined as the iteration of minimum validation loss, (2) test

accuracy and train accuracy at that checkpoint, and (3) the difference in test accuracy between the Winning-Ticket (WT) and Random-Reinitialization (RR) models. Models were trained with Adam at learning rate $3e-4$, batch size 128, validation checks every 100 iterations, a maximum of 5000 iterations per round, and early-stopping with a patience of 5 validation checks. Convolutional layers were pruned at 5% per layer and fully connected hidden layers at 10% globally each round. Results in this section are from a single training run.

Across the first six pruning rounds, both the WT and RR models held test accuracy within roughly two percentage points of the unpruned baseline. The unpruned CONV-4 reached 76.6% test accuracy with WT and 75.8% with RR. By the time the network was reduced to approximately 13% of its original parameter count (Conv 44/86, FC 33/180), test accuracy was still 73.6% for WT and 73.9% for RR, indicating that a substantial fraction of the original parameters in CONV-4 are not strictly necessary to maintain near-baseline classification quality on CIFAR-10. Train accuracy decreased steadily over the same range, from 91.2% to 83.0% for WT and 90.6% to 82.7% for RR, suggesting that pruning gradually reduced model capacity rather than triggering optimization collapse.

Below approximately 6% remaining weights, performance degraded sharply for both models. Test accuracy fell from the low-70s into the 40s within three rounds, and train accuracy dropped in lockstep. The early-stop iteration metric increased monotonically as the network shrank and effectively saturated at our 5000-iteration cap once the model reached the smallest sizes, indicating that very compressed CONV-4 architectures did not converge within our training budget rather than fitting quickly.

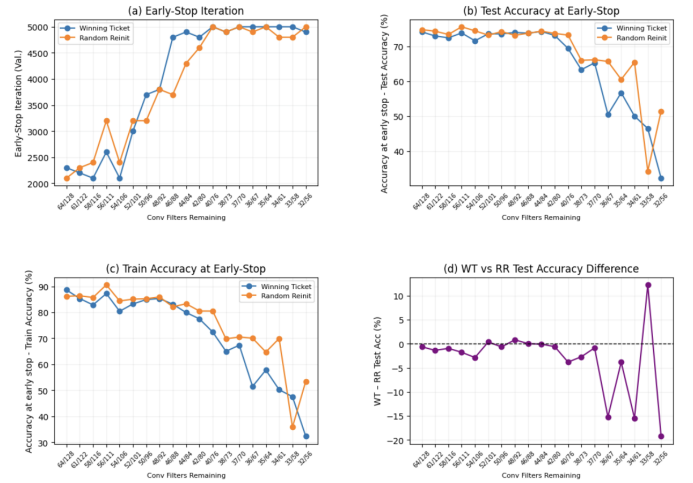


Fig. 4. CONV-4 Results

The WT versus RR comparison did not show a consistent advantage for the lottery-ticket initialization. Through the moderate pruning regime, the WT model held a small but unstable lead on test accuracy of roughly +1 percentage point. As pruning intensified, the gap oscillated in sign by as much

as ± 16 points across consecutive rounds before recovering to a small positive value at the smallest configuration tested. The single-run nature of this evaluation amplifies this noise, but the lack of stable monotonic separation is itself consistent with the WT vs RR oscillation observed in our CONV-2 experiments.

A distinctive characteristic of the CONV-4 hybrid pruning strategy is visible in the architecture-axis figure: the fully connected hidden layers collapsed asymmetrically under global ranking, with FC1 shrinking from 256 to 2 neurons across ten rounds while FC2 only reduced from 256 to 72. Under our outgoing-weight scoring metric, FC1 entries are systematically ranked lower because they each contribute to 256 outgoing connections versus only 10 for FC2 entries. This is a property of the pruning policy rather than an implementation artifact, and it ended pruning earlier than intended.

IV. CONCLUSION

A. LENET on MNIST

The MNIST experiments produced inconclusive results regarding the existence of structural lottery tickets in the fully connected setting. While the lottery ticket model showed a modest advantage in early stopping iteration during the early pruning rounds, consistent with faster convergence due to rewind initialization weights, this advantage diminished with further pruning and the accuracy curves for both models remained closely matched throughout. We were unable to establish a clear and consistent advantage for the lottery ticket initialization over random reinitialization at this scale.

We attribute the inconclusive outcome to two primary factors. First, MNIST is a relatively simple dataset that a small fully connected network can learn quickly regardless of initialization quality, which may have allowed the randomly initialized baseline to match the lottery ticket model without needing the advantage that rewind weights would otherwise provide. Second, computational constraints limited the number of runs and the thoroughness of hyperparameter exploration. The more meaningful test of the structural lottery ticket hypothesis is likely on CIFAR-10, where the learning task is harder and initialization quality is expected to matter more.

B. Conv-2 on CIFAR-10

Following the LENET conclusions, the existence of winning lottery tickets through our structured pruning methods were also inconclusive for the CONV-2 model within the CIFAR-10 dataset. Although the results showed promising signs after many rounds of structured pruning and resetting the weights back to their original initialization. In certain runs the accuracy was never as consistent. In addition, the advantages of Winning ticket initialization over Random Reinitialization that was presented in unstructured pruning in the LTH paper, were not as clearly seen in structured pruning. The Winning ticket did not have greater advantage over the number of iterations and at times had lower test accuracy. The promising signs ultimately represent that finding structured winning lottery tickets is much more difficult to find in comparison to unstructured pruning. The removal of neurons and filters adds more difficult

challenges in optimizing and finding such winning tickets as the structure becomes more limited.

C. CONV-4 on CIFAR-10

The CONV-4 experiments are consistent with the broader pattern reported in our LENET and CONV-2 sections: iterative structured pruning can reduce the size of a competitive CIFAR-10 model substantially without immediately collapsing test accuracy, but a clean, monotonic Winning-Ticket advantage over random reinitialization was not observed under our experimental setup. CONV-4 retained near-baseline test accuracy down to roughly 13% of its original parameter count, and accuracy collapsed only after the network entered very compressed regimes where the WT versus RR test-accuracy gap oscillated in sign rather than diverging in WT's favor.

Two findings stand out. Structural redundancy in CONV-4 was substantial during the moderate pruning phase, supporting the broader project conclusion that meaningful structured compression is achievable under iterative pruning. However, optimization difficulty grew with pruning depth: early-stop iteration rose monotonically and saturated at the 5000 iteration cap once the network became small, meaning highly compressed CONV-4 architectures need either a longer training budget or richer optimization to be evaluated fairly. The hybrid pruning policy also exposed an inherent imbalance in global FC ranking — FC1 collapsed roughly 36 times faster than FC2 due to the difference in outgoing-weight counts per neuron — biasing the late-stage architectures and ending pruning earlier than intended.

These results support the structural lottery-ticket hypothesis only weakly under our setup. CONV-4 demonstrates that the original initialization is sufficient to remain trainable after multiple rounds of iterative pruning, but it does not provide stable evidence that the WT initialization is consistently superior to a fresh random initialization at matched size. We therefore characterize the CONV-4 results as partially supportive but inconclusive. Future work should average over additional independent runs, increase the per-round training budget so deeply pruned networks can fully converge, and investigate normalization or layer-balanced variants of the global FC ranking rule to avoid the FC1 collapse seen here.

D. Final Remarks

Though our results were inconclusive, we believe that with more time and hardware we could reach meaningful outcomes. Smaller pruning rates, more rounds, and alternative importance scoring methods were all within reach, but fell outside the scope of our limited timeframe and resources.

REFERENCES

- [1] Frankle, J., & Carbin, M. (2019). *The lottery ticket hypothesis: Finding sparse, trainable neural networks*. arXiv. <https://arxiv.org/abs/1803.03635>
- [2] Liu, B., Zhang, Z., He, P., Wang, Z., Xiao, Y., Ye, R., Zhou, Y., Ku, W.-S., & Hui, B. (2024). *A survey of lottery ticket hypothesis*. arXiv. <https://arxiv.org/abs/2403.04861>